

Dark Matter Capture and Heating of White Dwarfs in Omega Centauri

Abstract

White dwarfs (WDs) can act as calorimeters for dark matter (DM) in sufficiently dense environments. Dark matter particles passing through a WD can scatter, become gravitationally captured, thermalize, and, for annihilating DM, inject energy into the star. This can prevent old WDs from cooling below a characteristic luminosity or effective-temperature floor. Omega Centauri is a particularly interesting target because it is widely interpreted as the remnant nucleus of an accreted dwarf galaxy and may therefore retain some fraction of its original dark halo. At the same time, recent stellar-dynamical studies show that Omega Centauri contains a substantial population of centrally concentrated stellar remnants, potentially together with an intermediate-mass black hole, complicating attempts to infer the surviving particle-DM distribution from central kinematics.

The deepest observed Omega Centauri WD cooling sequence, measured with HST by Scalco et al. (2024), lies at a projected radius of approximately 20 pc from the cluster centre and reaches the termination of the WD luminosity function at $m_{\text{F606W}} = 30.1 \pm 0.2$. The corresponding objects are old WDs with effective temperatures of order a few thousand kelvin and luminosities of order $10^{29} \text{ erg s}^{-1}$, although a quantitative conversion requires the appropriate WD cooling tracks and mass distribution. Existing kinematic and stripped-progenitor models suggest physically interesting DM densities at this radius of approximately 0.1 to a few $M_{\odot} \text{ pc}^{-3}$, or several to approximately 100 GeV cm^{-3} . The observed faint WD sequence therefore provides a potentially useful constraint on DM capture and heating at radii of order 20 pc, complementary to annihilation searches and central dynamical constraints.

1 Motivation: inelastic dark matter, the LZ event, and why white dwarfs

The original proposal that white dwarfs can capture and burn *inelastic* dark matter is due to McCullough and Fairbairn (2010, arXiv:1001.2737), followed by Hooper, Spolyar, Vallinotto and Gnedin (2010, arXiv:1002.0005), who showed that inelastic dark matter accumulating in compact stars can prevent old WDs from cooling and proposed WD populations in dwarf spheroidals and the Milky Way as a test.

The question has become topical. The LUX-ZEPLIN collaboration reports a single event in an extended nuclear-recoil window, with recoil energy $248 \pm 23 \text{ (stat)} \pm 23 \text{ (sys)} \text{ keV}$ and a global significance of 2.6σ (arXiv:2609.02823). Such a high-energy recoil is naturally produced by inelastic dark matter scattering with a mass splitting δ between two dark-sector states; McCabe (arXiv:2609.04181) shows that the preferred splitting sits near the kinematic limit, which predicts a strongly seasonal ($> 50\%$ annual modulation) signal.

The simplest candidate, a Higgsino, is disfavoured: dark matter captured by the Sun over its lifetime would annihilate and produce high-energy neutrinos, and their absence in IceCube and Super-Kamiokande requires $\delta \gtrsim 560 \text{ keV}$ (Pospelov and Ramani, arXiv:2609.02775; Bose et al., arXiv:2609.07807), incompatible with the LZ kinematics for standard halo velocities. The event may still be inelastic dark matter of a different dark-sector kind, for instance one annihilating to e^+e^- , which evades the solar-neutrino limits.

White dwarfs are the natural next probe because their compactness accelerates infalling dark matter to velocities that make inelastic up-scattering possible for mass splittings as large as ~ 5 MeV. They therefore cover the LZ-preferred region in a largely model-independent way, and the canonical Higgsino out to larger splittings. The constraint is only as good as the dark matter density the WDs sit in, which is why an independent, robust assessment of the dark matter content of Omega Centauri — the subject of the rest of this note and of the companion progenitor-orbit study — matters.

Open items collected for this programme:

- (i) *Theory/phenomenology*: equilibration timescales in WDs; benchmark heating luminosities for the LZ-motivated parameters (target: two weeks).
- (ii) *Exploitation*: given those benchmarks and the Omega Centauri WD observations, does the detailed dark matter distribution inside the cluster matter, or only an average?
- (iii) *Kinematics*: how the tidal features of Omega Centauri constrain its dark matter distribution.
- (iv) *Simulation*: a simplified late Omega Centauri–Milky Way encounter of a tidally stripped host that reproduces the present configuration, to see whether dark matter survives (Sections 9 and the companion note).

2 Dark matter capture and heating of white dwarfs

A dark matter particle passing through a white dwarf can scatter from nuclei, lose enough kinetic energy to become gravitationally bound, undergo subsequent scatterings, and eventually thermalize in the stellar interior. For annihilating dark matter, captured particles may subsequently annihilate and deposit energy into the star.

Once capture-annihilation equilibrium is reached, the annihilation luminosity is approximately determined by the capture rate,

$$L_\chi \sim m_\chi C, \quad (1)$$

up to factors describing the fraction of the annihilation energy that is deposited in the star rather than escaping in weakly interacting particles such as neutrinos.

The capture rate is generally a function of the ambient dark matter density and velocity distribution, the WD properties, and the microscopic particle physics:

$$C = C(\rho_\chi, f(v), M_{\text{WD}}, R_{\text{WD}}, \sigma_{\chi A}, m_\chi). \quad (2)$$

The corresponding WD luminosity is

$$L_{\text{WD}} = 4\pi R_{\text{WD}}^2 \sigma_{\text{SB}} T_{\text{eff}}^4. \quad (3)$$

If the dark matter heating becomes larger than the intrinsic cooling luminosity, then an old WD cannot cool below a characteristic temperature floor satisfying approximately

$$4\pi R_{\text{WD}}^2 \sigma_{\text{SB}} T_{\text{floor}}^4 \simeq L_\chi. \quad (4)$$

The cleanest observational signature is therefore not necessarily an anomalously luminous individual WD, but instead a modification of the faint end of the WD luminosity function. Depending on the dark matter model, one can obtain a shifted termination, a pile-up of objects near a heating floor, or the disappearance of WDs below a characteristic luminosity.

3 Foundational literature on WD capture and heating

Several early papers established compact stars, and WDs in particular, as probes of dark matter capture and heating.

Moskalenko and Wai (2007) introduced the idea of “dark matter burners,” in which capture and annihilation of dark matter in compact stars can provide an additional energy source.

Bertone and Fairbairn (2008) developed compact stars as dark matter probes and discussed how the high escape speeds and densities of WDs enhance capture.

McCullough and Fairbairn (2010) considered inelastic dark matter capture by WDs and emphasized that compact stars can probe regions of particle-physics parameter space inaccessible to terrestrial direct-detection experiments.

Hooper et al. (2010) similarly studied efficient capture and heating of compact stars, including the possibility of maintaining old WDs at effective temperatures of several thousand kelvin in sufficiently dense dark matter environments.

A useful modern review is Bramante and Raj (2024), which discusses dark matter capture, annihilation, accumulation, thermalization, collapse, and other compact-star probes.

For very heavy dark matter, capture becomes more complicated because multiple scatterings may be required before a particle loses sufficient energy to remain bound. Bell et al. (2024) showed that detailed treatment of trajectories, nuclear form factors, and multiple-scattering effects can substantially alter the capture rate relative to simplified estimates.

The basic observational comparison is therefore

$$L_\chi(\rho_\chi, f(v), m_\chi, \sigma_{\chi A}, M_{\text{WD}}, R_{\text{WD}}) \lesssim L_{\text{WD,obs}}, \quad (5)$$

for the faintest old WDs observed in the relevant environment.

4 An Omega Centauri-specific WD heating calculation

The most directly relevant Omega Centauri paper is Amaro-Seoane et al. (2016), which explicitly considered DM capture and annihilation in WDs in Omega Centauri and the possible effects of a dark matter cusp or “crest” associated with an intermediate-mass black hole (IMBH).

Their modelling began from an NFW-like progenitor halo and considered the effects of adiabatic contraction and subsequent stellar heating of the dark matter distribution.

In the absence of an IMBH, their model produced an approximately flattened dark matter distribution inside a characteristic radius of order

$$r_{\text{DMh}} \simeq 3.5 \text{ pc}. \quad (6)$$

With an IMBH, they considered a dark matter crest approximately of the form

$$\rho_\chi(r) \propto r^{-3/2} \quad (7)$$

inside the black-hole influence radius, which they took to be roughly

$$r_h \simeq 0.12 \text{ pc}. \quad (8)$$

Assuming capture near the geometrical saturation limit, they obtained characteristic minimum WD temperatures of approximately

$$T_{\text{eff,min}} \sim 7000 \text{ K} \quad (9)$$

in the cluster core.

In the IMBH crest scenario they obtained still higher temperatures, approximately

$$T_{\text{eff,min}} \sim 8500 \text{ K} \quad (10)$$

at $r \sim 0.1$ pc and

$$T_{\text{eff,min}} \sim 1.9 \times 10^4 \text{ K} \quad (11)$$

at $r \sim 0.01$ pc.

The important limitation was observational. At the time, photometric incompleteness and crowding in the central region of Omega Centauri prevented a reliable determination of whether cool WDs were genuinely absent. Therefore, the apparent lack of very cool central WDs could not be interpreted as a robust dark matter heating signal or constraint.

The newer, much deeper HST WD observations change the observational situation substantially, although the deepest field is located far outside the cluster centre.

5 The deepest Omega Centauri WD cooling sequence

The key observational result is Scalco et al. (2024), from the HST Large Programme on Omega Centauri. They measured the WD luminosity function sufficiently deeply to identify its faint-end turnover and termination.

The observed luminosity function peaks at

$$m_{\text{F606W}} = 30.1 \pm 0.2. \quad (12)$$

After completeness and field-contamination corrections, the WD counts decline rapidly beyond this point. Artificial-star tests continue to recover stars at comparable or fainter magnitudes, so the downturn is not simply the photometric detection limit.

The observed faint sequence is compatible with a predominantly old population, with the oldest component having an age of approximately 13 Gyr.

This observation is therefore potentially very useful for any mechanism that would impose an additional luminosity floor on old WDs.

6 Location of the deepest WDs in Omega Centauri

For dark matter heating, the position of the WD sample within the cluster is crucial.

Scalco et al. state that their deep field is located at approximately

$$R_{\text{proj}} = 2.6 r_h \quad (13)$$

from the centre of Omega Centauri.

Using a commonly adopted projected half-light radius

$$r_h \simeq 5.0', \quad (14)$$

this corresponds to

$$R_{\text{proj}} \simeq 13'. \quad (15)$$

At a distance of approximately

$$D \simeq 5.2 \text{ kpc}, \quad (16)$$

one arcminute corresponds to approximately

$$1' \simeq 1.51 \text{ pc}. \quad (17)$$

The projected physical radius of the deep field is therefore approximately

$$R_{\text{proj}} \simeq 19.7 \text{ pc} \quad (18)$$

or, to useful accuracy,

$$\boxed{R_{\text{proj}} \simeq 20 \text{ pc.}} \quad (19)$$

This is one of the most important facts for interpreting the data in terms of dark matter.

The deepest securely measured Omega Centauri cooling sequence is therefore not probing the central parsec, where the largest kinematically inferred “dark” densities occur. It is probing WDs at projected distances of order 20 pc.

Furthermore, because this is only the projected radius,

$$r_{3\text{D}} \geq R_{\text{proj}}, \quad (20)$$

so every WD in the field lies at an instantaneous three-dimensional cluster-centric radius of at least approximately 20 pc.

7 How cool are the faintest Omega Centauri WDs?

Scalco et al. do not assign a single effective temperature to the luminosity-function termination.

The faint end depends on the WD mass distribution and on detailed cooling physics. In particular, their theoretical cooling sequences show that the “blue hook” at the faint end is populated by relatively massive WDs,

$$M_{\text{WD}} \sim 1 - 1.1 M_{\odot}, \quad (21)$$

which formed relatively early and therefore have had almost the full cluster lifetime to cool.

Consequently, one should not convert the observed magnitude cutoff into a unique effective temperature using a canonical $0.6 M_{\odot}$ WD.

Nevertheless, the relevant physical regime is clearly a few thousand kelvin rather than approximately 8000 K.

For reference, for a canonical WD with

$$M_{\text{WD}} \simeq 0.6 M_{\odot} \quad (22)$$

and

$$R_{\text{WD}} \simeq 0.012 R_{\odot}, \quad (23)$$

the Stefan-Boltzmann relation gives approximately

$$T_{\text{eff}} = 3000 \text{ K} \quad \Rightarrow \quad L_{\text{WD}} \simeq 4 \times 10^{28} \text{ erg s}^{-1}, \quad (24)$$

$$T_{\text{eff}} = 4000 \text{ K} \quad \Rightarrow \quad L_{\text{WD}} \simeq 1.3 \times 10^{29} \text{ erg s}^{-1}, \quad (25)$$

$$T_{\text{eff}} = 5000 \text{ K} \quad \Rightarrow \quad L_{\text{WD}} \simeq 3 \times 10^{29} \text{ erg s}^{-1}. \quad (26)$$

Thus the luminosity scale relevant to the observed old WD sequence is approximately

$$\boxed{L_{\text{WD,faint}} \sim 10^{29} \text{ erg s}^{-1}} \quad (27)$$

to order of magnitude.

This number should not be interpreted as a precise luminosity measurement for every WD at the termination. A quantitative dark matter analysis should instead add the heating term directly to the appropriate WD cooling tracks, including the observed mass distribution, atmospheric properties, crystallization, and other relevant cooling physics.

However, a heating floor substantially larger than

$$L_{\chi} \sim 10^{30} \text{ erg s}^{-1} \quad (28)$$

would clearly be difficult to reconcile with the existence of the faint old WD sequence in this field.

8 Which dark matter density should be compared with the WD data?

The relevant dark matter density is not the central density inferred from kinematic modelling. The WDs defining the deepest cooling sequence are observed at

$$R_{\text{proj}} \sim 20 \text{ pc}, \quad (29)$$

so the appropriate comparison is with the dark matter distribution at radii of order 20 pc or with an orbit-averaged density along the stellar trajectories.

Several very different estimates of the Omega Centauri dark matter distribution exist in the literature.

8.1 Brown et al. (2019)

Brown et al. fitted stellar kinematics using an NFW-like dark matter component.

Their characteristic result was approximately

$$M_{\text{DM}}(< 7 \text{ pc}) \simeq 10^{5.3} M_{\odot} \quad (30)$$

with a characteristic NFW scale radius of approximately

$$r_s \simeq 1.6 \text{ pc}. \quad (31)$$

Taking the median fitted parameters literally and extrapolating the NFW profile outward gives a local density at 20 pc of order

$$\boxed{\rho_{\chi}(20 \text{ pc}) \sim 2 M_{\odot} \text{ pc}^{-3}}. \quad (32)$$

Using

$$1 M_{\odot} \text{ pc}^{-3} \simeq 37.97 \text{ GeV cm}^{-3}, \quad (33)$$

this corresponds to

$$\boxed{\rho_{\chi}(20 \text{ pc}) \sim 75 \text{ GeV cm}^{-3}}. \quad (34)$$

This value should be understood as an extrapolation from representative median NFW parameters rather than as a directly quoted posterior constraint on $\rho_{\chi}(20 \text{ pc})$.

The central densities of these fits are much larger, but the NFW scale radius is small, so the profile falls strongly by the time one reaches 20 pc.

8.2 Evans, Strigari and Zivick (2022)

Evans, Strigari and Zivick carried out more detailed axisymmetric dynamical modelling using several independent kinematic datasets.

Their HST-based solutions can favour an extremely compact dark component. Representative NFW parameters considered in their modelling have approximately

$$r_s \simeq 0.15 \text{ pc} \quad (35)$$

and

$$\rho_s \simeq 1.6 \times 10^6 M_{\odot} \text{ pc}^{-3}. \quad (36)$$

These parameters correspond to an extremely concentrated central profile.

However, because the scale radius is so small, extrapolation to approximately 20 pc gives a much lower density, of order

$$\boxed{\rho_{\chi}(20 \text{ pc}) \sim 0.6 M_{\odot} \text{ pc}^{-3}} \quad (37)$$

or

$$\boxed{\rho_\chi(20 \text{ pc}) \sim 23 \text{ GeV cm}^{-3}}. \quad (38)$$

Again, this should not be interpreted as a directly measured density at 20 pc. It is an extrapolation of a representative fitted NFW-like model.

A further complication is that the inferred central dark component depends substantially on the kinematic dataset. The HST measurements provide much of the preference for a compact central mass, while the outer Gaia kinematics show much less need for such a component.

8.3 Wirth, Bekki and Hayashi (2020)

The Wirth, Bekki and Hayashi simulations address a physically different question: whether a nuclear star cluster formed within a dwarf galaxy can retain dark matter after severe tidal stripping by the Milky Way.

Their surviving nuclear dark matter distributions typically have densities of approximately

$$\boxed{\rho_\chi \sim 0.1 - \text{few } M_\odot \text{ pc}^{-3}} \quad (39)$$

depending strongly on the initial halo, nuclear cluster formation site, progenitor orbit, and stripping history.

These densities are dramatically smaller than the very high central “dark” densities sometimes inferred from Jeans modelling.

However, at radii of order 20 pc, the physically motivated surviving-halo densities and the extrapolated Brown/Evans phenomenological profiles are intriguingly similar.

A reasonable first-pass density range to investigate for the deep Scalco field is therefore

$$\boxed{\rho_\chi \sim 0.1 - 3 M_\odot \text{ pc}^{-3}} \quad (40)$$

or approximately

$$\boxed{\rho_\chi \sim 4 - 110 \text{ GeV cm}^{-3}}. \quad (41)$$

This should be treated as a physically motivated range for exploration, not as a precise observational constraint.

9 Predicted dark matter density inside Omega Centauri from the progenitor-orbit analysis

The companion note ([progenitor_orbits.pdf](#)) reconstructs three classes of orbital history for the host dwarf — today’s orbit throughout (class 1), infall from the virial radius under dynamical friction (class 2), and deposition by Gaia–Sausage–Enceladus followed by bar-driven migration (class 3) — and walks Errani & Navarro (2021) tidal tracks along them. Two results carry over.

First, on every history a bare dark-matter halo is stripped far below the nucleus mass: a self-bound remnant would need a mean density $\simeq 20$ times the Milky Way’s mean density inside the pericentre, 1.7×10^{10} (1.6 kpc pericentres, classes 1–2) to $1.5 \times 10^{11} M_\odot \text{ kpc}^{-3}$ (0.4 kpc , class 3), which an NFW cusp of $10^{10} M_\odot$ reaches only inside a few parsecs. Any dark matter around Omega Centauri today is therefore held by the nucleus’s own potential.

Second, the budget is then the initial dark matter inside the nucleus’s Jacobi radius at the smallest pericentre, $r_J = r_{\text{peri}}[M_{\text{nuc}}/3M_{\text{MW}}(< r_{\text{peri}})]^{1/3}$: $33\text{--}36 \text{ pc}$ for class 3, $40\text{--}70 \text{ pc}$ for classes 1–2. For NFW haloes of $10^9\text{--}10^{11} M_\odot$ at $z = 2$ this is $2 \times 10^5\text{--}1.2 \times 10^6 M_\odot$ (class 3) and $0.3\text{--}4.4 \times 10^6 M_\odot$ (classes 1–2).

9.1 The density profile if the dark matter inside r_J is retained

Figure 1 shows the simplest model consistent with these results: the initial NFW cusp, unchanged inside r_J and exponentially truncated there,

$$\rho_{\text{DM}}(r) = \rho_{\text{NFW}}(r; M_{200}, c) e^{-r/r_J}, \quad (42)$$

for $M_{200} = 10^9, 10^{10}, 10^{11} M_\odot$, concentrations $c = 5, 10, 15$ and the two Jacobi radii, against the stellar density of the cluster (the project’s MGE light model scaled to the fitted stellar mass $2.9 \times 10^6 M_\odot$). The three radii of interest are marked: 3 pc (where central kinematics and stellar remnants dominate), 10 pc, and 20 pc, the projected radius of the deep WD field.

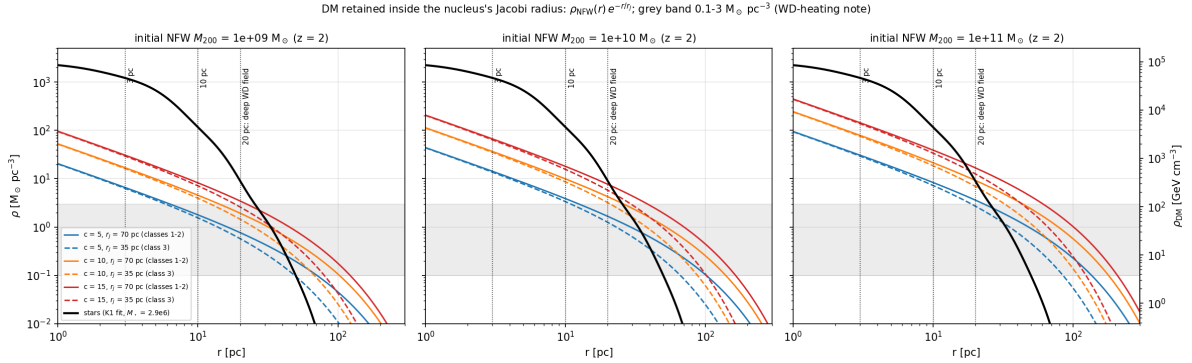


Figure 1: Dark matter retained inside the nucleus’s Jacobi radius, $\rho_{\text{NFW}}(r) e^{-r/r_J}$, for initial NFW haloes at $z = 2$ (panels: M_{200} ; colours: concentration; solid: $r_J = 70$ pc, classes 1–2; dashed: $r_J = 35$ pc, class 3), compared with the stellar density (black). Dotted lines mark 3, 10 and 20 pc; the grey band is the $0.1\text{--}3 M_\odot \text{ pc}^{-3}$ range discussed in Section 8. Right axis in GeV cm^{-3} .

Table 1: Retained dark matter density [$M_\odot \text{ pc}^{-3}$] at the three radii for $r_J = 35$ pc (class 3) / 70 pc (classes 1–2); stellar density 1230, 117, $9.1 M_\odot \text{ pc}^{-3}$ at 3, 10, 20 pc. Multiply by 38 for GeV cm^{-3} .

initial halo	3 pc	10 pc	20 pc
$10^9, c = 5$	6.3 / 6.6	1.6 / 1.8	0.58 / 0.77
$10^9, c = 15$	30 / 31	7.1 / 8.2	2.6 / 3.5
$10^{10}, c = 5$	14 / 14	3.4 / 3.9	1.3 / 1.7
$10^{10}, c = 10$	35 / 37	8.6 / 9.9	3.2 / 4.3
$10^{10}, c = 15$	64 / 67	16 / 18	5.8 / 7.7
$10^{11}, c = 5$	30 / 31	7.2 / 8.4	2.7 / 3.6
$10^{11}, c = 15$	139 / 145	34 / 39	13 / 17

Three things follow. (i) Inside r_J the density is the initial cusp, $\rho \simeq \rho_s r_s / r \propto M_{200}^{1/3}$ times a concentration factor; the Jacobi radius, and therefore the class of orbital history, changes the density at 20 pc by at most a factor of two, whereas halo mass and concentration each span a factor of five. The orbital class controls the retained *mass* (through r_J^3) much more than the local *density* a WD sees. (ii) At 20 pc the grid spans $0.6\text{--}17 M_\odot \text{ pc}^{-3}$ ($23\text{--}640 \text{ GeV cm}^{-3}$) against $9 M_\odot \text{ pc}^{-3}$ in stars: dark matter comparable to or above the stars for $M_{200} \geq 10^{10}$ with $c \geq 10$, or any 10^{11} halo. The $0.1\text{--}3 M_\odot \text{ pc}^{-3}$ range adopted in Section 8 covers only the low-mass, low-concentration corner; the physically plausible grid mostly predicts $1\text{--}17 M_\odot \text{ pc}^{-3}$, so the capture calculation should be evaluated at $\rho_\chi \sim 10 M_\odot \text{ pc}^{-3} \simeq 400 \text{ GeV cm}^{-3}$ as well. (iii) At

3 pc the stars dominate by 10–200, at 10 pc by 3–70; the 20 pc field is where an extended dark component is most detectable, kinematically as well as calorimetrically.

9.2 What the frozen-cusp model leaves out

The profile inside r_J is an assumption, not a result, and it is optimistic beyond ~ 10 pc for one reason and uncertain inside for two others. Pericentric tidal shocks act on material inside r_J as well; over ~ 100 passages (class 3 pericentres at 0.4 kpc) they erode the outer part of the retained dark matter, which can only lower the profile near r_J . The dark matter was not a pristine cusp when stripping began: the nucleus’s formation adiabatically contracted it (raising the inner densities), while baryonic feedback in the host may have cored it (lowering everything and making complete stripping possible). And inside a few parsecs two-body heating by stars and remnants diffuses the dark matter outward over 10 Gyr. The first effect is what the planned N -body runs quantify; the second is an initial-condition choice to be bracketed.

10 The central kinematic “dark” component is not necessarily particle dark matter

A central complication is that the dynamically inferred non-luminous mass in Omega Centauri can be naturally explained, at least in substantial part, by compact stellar remnants.

Recent multi-mass dynamical modelling indicates that Omega Centauri may retain approximately a few $10^5 M_\odot$ in centrally concentrated stellar remnants, particularly stellar-mass black holes.

Banares-Hernandez et al. infer an extended central dark component of approximately

$$M_{\text{rem}} \sim (2 - 3) \times 10^5 M_\odot \quad (43)$$

with a characteristic spatial scale of approximately

$$a \sim 1.5 - 2.2 \text{ pc}. \quad (44)$$

For a Plummer-like distribution, the central density is

$$\rho_0 = \frac{3M}{4\pi a^3}, \quad (45)$$

giving characteristic densities of order

$$\rho_0 \sim 5 \times 10^3 - 2 \times 10^4 M_\odot \text{ pc}^{-3}. \quad (46)$$

Thus a dense stellar-remnant population can naturally produce precisely the kind of central non-luminous mass that could be absorbed into a phenomenological “DM” component if stellar remnants, an IMBH, and particle dark matter are not modelled separately.

A useful schematic decomposition is

$$\Phi(r) = \Phi_\star(r) + \Phi_{\text{rem}}(r) - \frac{GM_\bullet}{r} + \Phi_{\text{DM}}(r). \quad (47)$$

The physically relevant spatial scales can be very different:

$$\text{IMBH :} \quad r \lesssim 0.1 \text{ pc}, \quad (48)$$

$$\text{stellar remnants :} \quad r \sim 0.1 - \text{few pc}, \quad (49)$$

$$\text{surviving DM halo :} \quad r \sim \text{few} - 100 \text{ pc}. \quad (50)$$

The fact that the deepest WD field lies at approximately 20 pc is therefore potentially advantageous. At this radius, the influence of a highly concentrated remnant population is much smaller, whereas a genuinely extended surviving particle-DM halo may still contribute appreciably.

11 Orbital averaging of the WD dark matter exposure

A WD observed at a projected radius of 20 pc does not remain at 20 pc throughout its lifetime.

The internal orbital period of a star in Omega Centauri is short compared with the WD cooling age. A characteristic estimate is

$$T_{\text{orb}} \sim \frac{2\pi r}{v_c}. \quad (51)$$

For

$$r \sim 20 \text{ pc} \quad (52)$$

and

$$v_c \sim 20 - 30 \text{ km s}^{-1}, \quad (53)$$

the orbital period is only of order

$$T_{\text{orb}} \sim \text{few Myr}. \quad (54)$$

An old WD with a cooling age of many Gyr therefore executes thousands of internal orbits.

Consequently, the relevant capture rate is not necessarily determined by the instantaneous density at the observed projected radius.

The instantaneous capture rate is schematically

$$C(t) \propto \rho_\chi[r(t)] \mathcal{F}[v_{\text{rel}}(t)], \quad (55)$$

where the function $\mathcal{F}$ includes the velocity dependence of the gravitational focusing and scattering/capture probability.

If the dark matter capture-annihilation system and the stellar thermal response track changes in the ambient density rapidly compared with the stellar orbital period, then the heating can vary along the orbit.

If the relevant equilibration or thermal timescales are long compared with the orbital period, the effective quantity becomes closer to an orbital average,

$$\langle C \rangle_{\text{orb}} = \frac{1}{T_{\text{orb}}} \int_0^{T_{\text{orb}}} C[t] dt. \quad (56)$$

Schematically,

$$\langle C \rangle_{\text{orb}} \propto \langle \rho_\chi[r(t)] F[v_{\text{rel}}(t)] \rangle. \quad (57)$$

This is potentially important for centrally concentrated DM profiles.

A WD observed at 20 pc may have a considerably smaller pericentre and periodically pass through regions with much higher dark matter density.

Therefore the relevant quantity for a quantitative WD heating analysis is not simply

$$L_\chi[\rho_\chi(20 \text{ pc})], \quad (58)$$

but rather the capture and heating rate integrated over dynamically allowed stellar orbits.

12 What can the current Omega Centauri WD data constrain?

The direct observational statement is the following.

At a projected cluster-centric radius of approximately

$$R_{\text{proj}} \simeq 20 \text{ pc}, \quad (59)$$

Omega Centauri contains an old WD population whose luminosity function reaches its faint termination at

$$m_{\text{F606W}} = 30.1 \pm 0.2. \quad (60)$$

There is therefore no evidence in this field for a strong exotic heating floor that prevents the oldest WDs from reaching the observed faint luminosities.

For any particular particle-DM model, one can therefore require schematically

$$L_\chi(\rho_\chi, \sigma_{\chi N}, m_\chi, f(v), \dots) \lesssim L_{\text{faint WD}}. \quad (61)$$

A characteristic scale is

$$L_{\text{faint WD}} \sim 10^{29} \text{ erg s}^{-1}, \quad (62)$$

although a rigorous constraint must use the actual cooling tracks.

The current deep data therefore potentially constrain DM capture and heating at radii of order 20 pc.

They do *not*, however, directly test the very large central densities inferred in some phenomenological dynamical models at

$$r \lesssim 1 \text{ pc}. \quad (63)$$

The DM-heated WD temperatures of approximately 7000–19000 K discussed by Amaro-Seoane et al. were predominantly predictions for the central few parsecs and, in the strongest IMBH-crest cases, for radii below approximately 0.1 pc.

The deep Scalco field is much farther out.

13 A radial WD test of the Omega Centauri dark matter profile

There is therefore a particularly clean observational strategy.

At present, the situation is approximately

$$r \sim 20 \text{ pc} : \quad \text{deep WD luminosity function, moderate expected DM density}, \quad (64)$$

whereas

$$r \lesssim 1\text{--}5 \text{ pc} : \quad \text{potentially much higher DM density, but no comparably deep WD cooling sequence.} \quad (65)$$

If deep and complete WD luminosity functions could be obtained in several radial bins, for example

$$R \simeq 1, 3, 5, 10, 20 \text{ pc}, \quad (66)$$

then DM heating would predict a characteristic radial dependence in the cooling floor.

In a regime where capture is saturated and other WD properties are fixed approximately,

$$L_\chi \propto \rho_\chi, \quad (67)$$

which would imply

$$T_{\text{floor}} \propto L_\chi^{1/4} \propto \rho_\chi^{1/4}. \quad (68)$$

Ordinary WD cooling physics should not naturally produce a radial temperature floor that tracks an independently inferred dark matter profile.

A radial WD luminosity-function study could therefore provide a powerful and complementary diagnostic of the presence or absence of a surviving extended dark halo.

14 Complementarity with other Omega Centauri dark matter probes

The WD heating observable probes a different aspect of the dark matter distribution from conventional indirect searches.

For annihilation searches,

$$J = \int d\Omega \int_{\text{los}} \rho_\chi^2 ds. \quad (69)$$

The annihilation signal is therefore extremely sensitive to the unresolved inner density profile and to whether the central non-luminous mass is particle DM, stellar remnants, or an IMBH-related structure.

By contrast, WD capture is approximately linear in the local dark matter density,

$$C \propto \rho_\chi, \quad (70)$$

modulo the velocity-distribution and particle-physics dependence.

The two observables therefore probe quite different quantities.

A WD population at approximately 20 pc can constrain an extended DM component even if the central annihilation J -factor is highly uncertain.

Likewise, stellar kinematics primarily constrain the gravitational potential or enclosed mass,

$$M(< r), \quad (71)$$

and cannot by themselves cleanly determine how much of the unseen mass consists of stellar remnants rather than particle DM.

A combined analysis could therefore use:

- (i) inner stellar and pulsar kinematics to constrain the gravitational potential;
- (ii) tidal tails and extended debris to constrain mass at tens of parsecs and the progenitor stripping history;
- (iii) WD cooling to constrain local or orbit-averaged DM capture and heating;
- (iv) annihilation or decay searches to constrain particle-DM emission.

15 A useful first-pass parameter range

For the deepest currently observed WD population, the useful order-of-magnitude summary is

$$R_{\text{deepest WD}} \simeq 20 \text{ pc}, \quad (72)$$

$$T_{\text{coolest}} \sim \text{few} \times 10^3 \text{ K}, \quad (73)$$

$$L_{\text{coolest}} \sim 10^{29} \text{ erg s}^{-1} \quad (\text{approximate}), \quad (74)$$

$$\rho_{\chi, \text{plausible}}(20 \text{ pc}) \sim 0.1 - \text{few } M_\odot \text{ pc}^{-3}, \quad (75)$$

$$\rho_{\chi, \text{plausible}}(20 \text{ pc}) \sim 4 - 100 \text{ GeV cm}^{-3}. \quad (76)$$

The density range is motivated by a combination of stripped-progenitor simulations and extrapolations of phenomenological kinematic fits.

It should not be treated as a measured posterior interval.

16 A natural next calculation

A quantitative Omega Centauri WD dark matter analysis could proceed as follows.

First, adopt one or more physically motivated present-day DM distributions,

$$\rho_\chi(r), \quad (77)$$

for example:

- (i) the Brown et al. phenomenological NFW solution;
- (ii) representative Evans et al. compact NFW solutions;
- (iii) surviving profiles from the Wirth et al. stripped-nucleus simulations;
- (iv) more general tidally truncated cusped or cored profiles.

Second, construct dynamically consistent WD orbits within the observed Omega Centauri potential.

For each orbit, calculate

$$C(t) = C[\rho_\chi(r(t)), v_{\text{rel}}(t), m_\chi, \sigma_{\chi N}, M_{\text{WD}}, R_{\text{WD}}]. \quad (78)$$

Then determine the appropriate time-dependent or orbit-averaged heating rate,

$$L_\chi(t) \quad (79)$$

or

$$\langle L_\chi \rangle_{\text{orb}}, \quad (80)$$

depending on the capture-annihilation equilibration and WD thermal response timescales.

The heating term should then be inserted into the same cooling tracks used to model the Scalco et al. WD population.

Rather than comparing only to an individual faint WD, the most powerful observable would likely be the full luminosity function,

$$\frac{dN_{\text{WD}}}{dM_{\text{F606W}}}, \quad (81)$$

including its peak, width, descending branch, and termination.

The likelihood would then constrain the particle-physics parameters,

$$p(m_\chi, \sigma_{\chi N} \mid \text{WD LF}, \rho_\chi(r), \text{dynamics}), \quad (82)$$

or, conversely, could constrain the allowed DM density for a fixed particle model.

A more complete treatment would marginalize simultaneously over the uncertain DM distribution:

$$p(m_\chi, \sigma_{\chi N} \mid \text{data}) = \int p(m_\chi, \sigma_{\chi N} \mid \text{data}, \theta_{\text{DM}}) p(\theta_{\text{DM}} \mid \text{dynamics}) d\theta_{\text{DM}}. \quad (83)$$

17 Main conclusions

The present literature suggests the following picture.

First, dark matter capture and annihilation in WDs can impose an observable luminosity and effective-temperature floor in sufficiently dense dark matter environments.

Second, this mechanism has already been studied specifically for Omega Centauri. Amaro-Seoane et al. predicted central WD temperature floors ranging from approximately 7000 K to

nearly 2×10^4 K in models containing a dense central DM distribution and, in some cases, an IMBH-induced crest.

Third, the old central photometry available at that time was too incomplete to provide a robust test.

Fourth, Scalco et al. (2024) have now detected the termination of the Omega Centauri WD luminosity function at

$$m_{F606W} = 30.1 \pm 0.2, \quad (84)$$

providing a population of very old, faint WDs with characteristic temperatures of order a few thousand kelvin.

Fifth, these deepest WDs are located at a projected radius of approximately

$$R_{\text{proj}} \simeq 20 \text{ pc}, \quad (85)$$

rather than in the central parsec.

Sixth, physically plausible surviving-DM models and extrapolated phenomenological kinematic fits suggest DM densities at this radius of approximately

$$\rho_{\chi}(20 \text{ pc}) \sim 0.1 - \text{few } M_{\odot} \text{ pc}^{-3} \quad (86)$$

or

$$\rho_{\chi}(20 \text{ pc}) \sim 4 - 100 \text{ GeV cm}^{-3}. \quad (87)$$

Seventh, the current WD observations therefore provide a potentially interesting calorimetric constraint on an extended surviving Omega Centauri dark halo at radii of order 20 pc.

Finally, the appropriate comparison should ultimately use the orbit-averaged dark matter capture rate rather than simply the instantaneous density at the observed projected radius, because old WDs complete thousands of internal Omega Centauri orbits over their cooling lifetimes.

The most compelling future test would be a radially resolved WD cooling sequence extending from the central few parsecs to approximately 20 pc. Such a measurement could search directly for the radial heating pattern expected from a surviving dark matter halo while remaining complementary to central stellar dynamics, tidal-tail constraints, and conventional indirect-detection searches.

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